

Leveraging AI for Maritime Pattern of Life (POL) Analysis and Predictive Operations

Project Team

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Chapter 1

1 Introduction

This document outlines the development of a **Predictive System for Commercial Shipping Operations**. The system will analyze historical datasets to provide trends and predictive insights that will assist in decision-making processes relating to deployment frequencies, operational statuses, maintenance activities, and other factors critical to managing large commercial shipping fleets. The system will leverage machine learning models, the system will deliver forecasts that help optimize fleet management, improve operational efficiency, and reduce costs.

The document is intended for several types of readers:

- **Developers:** For them to understand the technical specifications and implementation details of the system.
- **Project Managers:** For them to track the project's progress and ensure that the defined scope aligns with business needs.
- **Marketing Staff:** From them to identify key benefits and marketable features of the system.
- **Testers:** From them to evaluate the system's functionality, accuracy, and performance.
- **Documentation Writers:** To create user manuals and technical documents based on the provided requirements and features.

1.1 Problem Statement

Fleet operations management in commercial shipping poses a big challenge for efficient management. In the commercial shipping industry, fleet managers often struggle to predict when vessels should be deployed, maintained, or just kept running to avoid unplanned downtime, inadequate resource allocation, and increase in costs relating to operational expenses. In the absence of predictive tools, fleet managers and operators rely too much on manual oversight which is based on too much guessing and implies an error-prone approach that makes little use of the mountains of historical data available to improve the decisions being made at all levels.

The proposed system addresses these challenges by utilizing historical datasets from commercial shipping operations. This system analyzes trends of vessel deployment and maintenance activities and operational statuses with generation of predictive insights for the fleet managers. These insights will allow fleet managers to anticipate future needs, optimize resource allocation, and prevent unexpected disruptions. Thus, by reducing uncertainty, the system aims to streamline operations and improve overall fleet efficiency.

1.2 Scope

The **Predictive System for Commercial Shipping Operations** is designed to provide data-driven insights into fleet management by analyzing historical datasets. The system's scope encompasses several key functionalities:

- **Data Collection and Integration:** The system collects historical data from various sources, including AIS (Automatic Identification System), radar logs, and maintenance records. This data is then preprocessed, cleaned, and structured for analysis.
- **Pattern Recognition and Correlation Analysis:** The system identifies patterns and correlations within the data, such as the frequency of deployments, correlations between different vessel types, and operational trends. This helps improve decision-making capabilities.
- **Predictive Analytics:** The system forecasts various operational metrics, including deployment patterns, future operational statuses, expected maintenance schedules, and potential surges in vessel deployments. These predictions enable more effective fleet management and help prevent costly disruptions.
- **User Interface and Visualization:** The system offers an intuitive, web-based user interface that presents predictive insights through reports, alerts, and visualizations. Fleet managers can easily access the system's recommendations, allowing them to take informed actions.
- **System Testing and Evaluation:** The system will be tested extensively using historical datasets to ensure that its predictions are accurate and actionable. Testing will confirm the system's ability to support the fleet management process effectively.

The scope of this project is limited to providing a proof-of-concept system that demonstrates the feasibility and potential benefits of predictive analytics in commercial shipping. It will not include live integration with existing operational systems but will lay the groundwork for future expansion.

1.3 Modules/Action Plan

The system will be based on a modular approach where there will be incremental development as well as the testing of critical features during the design phase. Every module is a different functionality that is going to get integrated to form the overall working of the system:

2.2.1 Data Integration and Preprocessing Module

- **Description:** This module will ingest historical data from many sources like AIS, Radar, and maintenance logs. The ingested data will be cleaned, normalized, and structured so that it is prepared for analysis.
- **Key Tasks:**
 - Develop methods for data ingestion and preprocessing.
 - Data cleaning techniques to handle missing or inconsistent data.
 - Unified structure for all incoming data to facilitate efficient processing.

2.2.2 Pattern Recognition Module

- **Description:** This module applies machine learning algorithms to analyze historical data to identify any dominant patterns of vessel deployment, maintenance, and operational statuses.
- **Key Tasks:**
 - Implement clustering and classification algorithms to group similar vessel types and their behaviors.
 - Identify the patterns regarding deployment based on historical data to make potential predictions on future trends.
 - Draw visualizations on typical trends in the commercial shipping space.

2.2.3 Predictive Analytics Module

- **Description:** The predictive analytics module employs advanced algorithms to build the future release of the vessel, maintenance requirements, and operational status from previous history.
- **Key Tasks:**
 - Design predictive models for predicting future call backs.
 - Create maintenance forecasting models to predict upcoming maintenance schedules.
 - Integrate predictive real-time updates and responses on upcoming trends for fleet managers.

2.2.4 Anomaly Detection Module

- **Description:** This module will involve anomaly detection of operational patterns, such as sudden increases in maintenance needs or unexplained changes in the vessel's behavior.
- **Key Tasks:**
 - Implement algorithms to flag anomalies in operational and maintenance patterns.
 - Design an alert mechanism to communicate the message to the operators when unusual behavior is detected.

2.2.5 User Interface and Visualization Module

- **Description:** The user interface of the system will allow the fleet managers and operators to visualize the predictive insights generated by the system.
- **Key Tasks:**
 - Design and develop a web interface that renders reports, predictions, alerts, etc.
 - Incorporate interactive visualizations like charts, graphs, and maps.
 - Develop a notification and alert management system to provide actionable insights.

2.2.6 System Integration and Testing Module

- **Description:** This module shall cover all the above modules and will integrate all as one system and performs detailed testing for the validation of accuracy and performance of the system.

- **Key Tasks:**
 - Develop integration strategies for combining data, predictive models, and the user interface.
 - Create detailed test plans to evaluate system accuracy using historical datasets.
 - Conduct performance testing and optimize system components where needed.

1.4 User Classes and Characteristics

The system is designed for several key user classes, each with specific roles and requirements.

Table 1. User Classes and Characteristics

User Classes	Description
Fleet Manager	A Fleet Manager is responsible for overseeing vessel operations, scheduling deployments, and maintaining optimal fleet performance. They will use the system to access predictive insights, optimize resource allocation, and improve decision-making in operational tasks.
Maintenance Team	The Maintenance Team ensures that vessels are well-maintained and operational. The system will provide them with predictive maintenance alerts, allowing them to schedule repairs and inspections proactively.
System Administrator	The System Administrator will manage the setup, data integration, and maintenance of the system. They will ensure the system's smooth operation and handle any technical configurations.
Commercial Shipping Companies	Represented by the Fleet Managers and Maintenance Teams, Commercial Shipping Companies are the primary beneficiaries of the system. Their interest lies in optimizing overall efficiency and reducing operational costs through the system's insights.

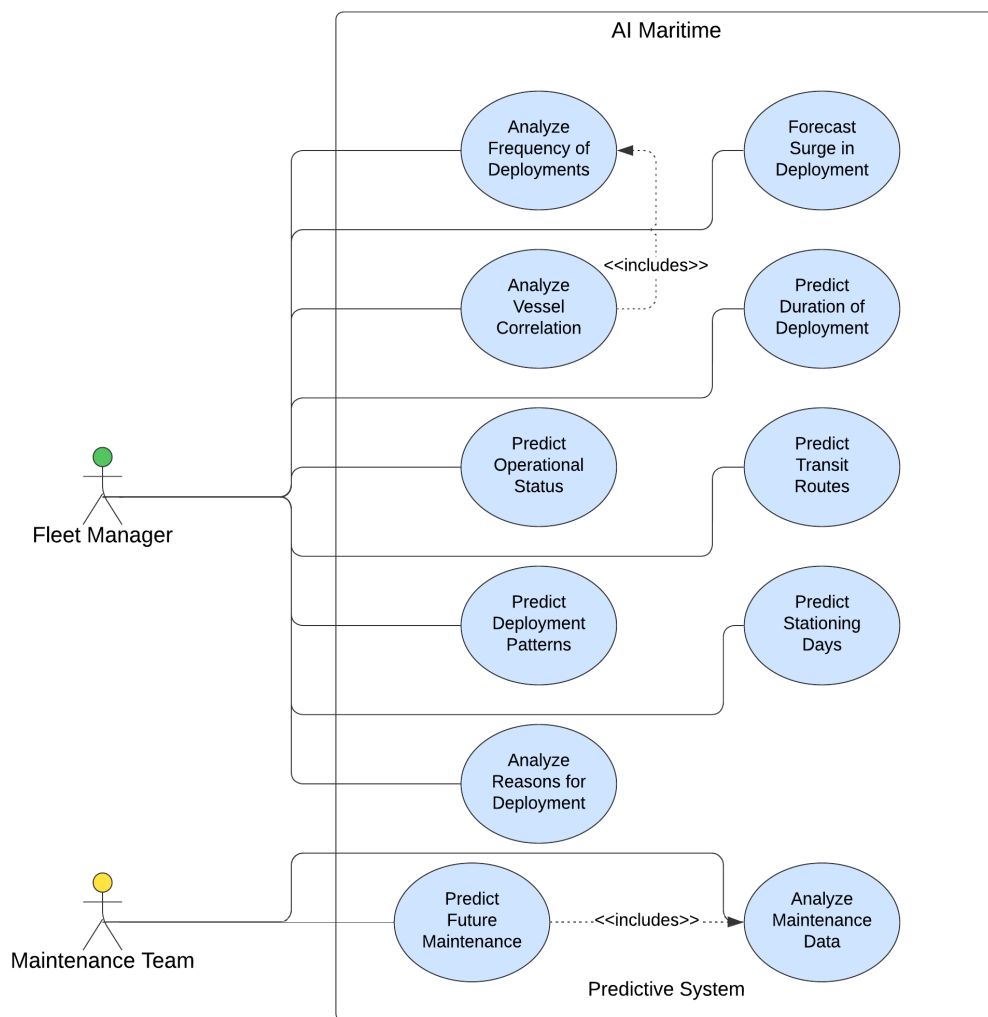
Chapter 2

2 Project Requirements

2.1 Use Cases

2.1.1 Use Case Diagram

Figure 1. Use Case Diagram



2.1.2 High Level/Brief Use Cases

UC-01

Name	Analyze Frequency of Deployments
Actor	Fleet Manager, Predictive System
Type	Primary
Description	The system analyzes historical data to identify patterns in the frequency of vessel deployments and predicts future deployment trends based on this data.

UC-02

Name	Analyze Vessel Correlations
Actor	Fleet Manager, Predictive System
Type	Primary
Description	The system identifies correlations between different types of vessels and their deployment frequencies, providing insights into vessel usage patterns.

UC-03

Name	Predict Operational Status
Actor	Fleet Manager, Predictive System
Type	Primary
Description	The system predicts future operational statuses of vessels by analyzing historical data and current trends, helping fleet managers anticipate operational needs.

UC-04

Name	Predict Deployment Patterns
Actor	Fleet Manager, Predictive System
Type	Primary
Description	The system predicts future deployment patterns by analyzing historical data on deployment during specific times, such as fishing seasons or holidays.

UC-05

Name	Analyze Reasons for Deployment
Actor	Fleet Manager, Predictive System
Type	Primary
Description	The system analyzes possible reasons for vessel deployment and identifies next areas of deployment, using correlations with factors such as vessel type and region.

UC-06

Name	Predict Stationing Days
Actor	Fleet Manager, Predictive System
Type	Primary
Description	The system predicts how many days vessels will be stationed at specific locations by analyzing past stationing data and correlating it with external factors like weather and vessel type.

UC-07

Name	Forecast Surge in Deployment
Actor	Fleet Manager, Predictive System
Type	Primary
Description	The system analyzes historical deployment data to predict future surges in vessel deployments based on trends and correlations with factors like weather or demand.

UC-08

Name	Predict Transit Routes
Actor	Fleet Manager, Predictive System
Type	Primary
Description	The system predicts probable transit routes based on historical data on vessel movements and factors like weather and sea conditions.

UC-09

Name	Predict Duration of Deployment
Actor	Fleet Manager, Predictive System
Type	Primary
Description	The system predicts the expected duration of vessel deployments at specific locations by analyzing historical data and external factors.

UC-10

Name	Analyze Maintenance Data
Actor	Maintenance Team, Predictive System
Type	Primary
Description	The system analyzes historical maintenance activity data to identify patterns and correlations with factors like weather, vessel type, and operational status.

UC-11

Name	Predict Future Maintenance
Actor	Maintenance Team, Predictive System
Type	Primary
Description	The system predicts future maintenance needs for vessels by analyzing past maintenance data and identifying correlations with factors like vessel usage and weather.

2.1.3 Fully Dressed/Expanded Use Cases

Table 2. Analyze Frequency of Deployments

UC-01		
Use Case Name	Analyze Frequency of Deployments	
Scope	Predictive System for Commercial Shipping	
Level	High-Level	
Primary Actor	Fleet Manager	
Stakeholder & Interest	Fleet Manager; interested in understanding deployment patterns for operational efficiency.	
Pre-Conditions	The system has historical deployment data available for analysis.	
Post-Conditions	The system provides insights into deployment frequency patterns and future deployment predictions.	
Main Success Scenario	User Action	System Response
	1. The Fleet Manager requests an analysis of deployment frequency.	2. The system retrieves historical deployment data from the database.
	3. The Fleet Manager specifies a time range for the analysis.	4. The system filters the data based on the specified time range.
	5. The Fleet Manager initiates the analysis.	
		6. The system analyzes the deployment data to identify patterns.
		7. The system presents the deployment frequency patterns in a report.
	8. The Fleet Manager views the results.	
Extensions	If no historical data is available, the system alerts the user and suggests additional data sources.	

Table 3. Analyze Vessel Correlation

UC-02	
Use Case Name	Analyze Vessel Correlations
Scope	Predictive System for Commercial Shipping
Level	High-Level
Primary Actor	Fleet Manager
Stakeholder & Interest	Fleet Manager; interested in understanding vessel usage patterns for resource allocation.
Pre-Conditions	The system has historical vessel deployment data available for analysis.
Post-Conditions	The system identifies correlations between vessel types and their deployment frequencies.
Main Success Scenario	User ActionSystem Response
	1. The Fleet Manager requests vessel correlation analysis.2. The system retrieves historical vessel deployment data from the database.
	3. The Fleet Manager specifies vessel types to analyze.4. The system filters the data based on the specified vessel types.
	5. The Fleet Manager initiates the analysis.6. The system analyzes the data to identify correlations between vessel types and their deployment frequencies.
	7. The system presents the correlations in a detailed report.
	8. The Fleet Manager views the results.
Extensions	If correlations are not significant, the system alerts the user with suggestions for further analysis.

Table 4. Predict Operational Status

UC-03	
Use Case Name	Predict Operational Status
Scope	Predictive System for Commercial Shipping
Level	High-Level
Primary Actor	Fleet Manager
Stakeholder & Interest	Fleet Manager; interested in forecasting operational needs to avoid downtime.
Pre-Conditions	The system has current and historical operational data available.
Post-Conditions	The system provides predictions for future operational statuses of vessels.
Main Success Scenario	User Action
	1. The Fleet Manager requests operational status predictions.
	3. The Fleet Manager selects the time frame for predictions.
	5. The Fleet Manager initiates the prediction process.
	8. The Fleet Manager views the predicted statuses.
	System Response
	2. The system retrieves relevant operational data from the database.
	4. The system filters the data based on the selected time frame.
	6. The system analyzes the data and forecasts future operational statuses.
	7. The system presents the correlations in a detailed report.
Extensions	If predictions indicate a high likelihood of operational issues, the system alerts the Fleet Manager with recommended actions.

Table 5. Predict Deployment Patterns

UC-04		
Use Case Name	Predict Deployment Patterns	
Scope	Predictive System for Commercial Shipping	
Level	High-Level	
Primary Actor	Fleet Manager	
Stakeholder & Interest	Fleet Manager; interested in forecasting future deployments for resource planning.	
Pre-Conditions	The system has access to historical deployment data.	
Post-Conditions	The system generates predictions for future deployment patterns.	
Main Success Scenario	User Action	System Response
	1. The Fleet Manager requests future deployment predictions.	2. The system retrieves historical deployment data from the database.
	3. The Fleet Manager specifies relevant factors (e.g., season).	4. The system analyzes the specified factors along with historical data.
	5. The Fleet Manager initiates the prediction process.	6. The system forecasts future deployment patterns based on the analysis.
		7. The system displays the predictions in an easy-to-read report format.
	8. The Fleet Manager views the predicted patterns.	
Extensions	If predictions are uncertain, the system offers alternative scenarios based on different variables.	

Table 6. Analyze Reasons for Deployment

UC-05		
Use Case Name	Analyze Reasons for Deployment	
Scope	Predictive System for Commercial Shipping	
Level	High-Level	
Primary Actor	Fleet Manager	
Stakeholder & Interest	Fleet Manager; interested in understanding the rationale behind vessel deployments for strategic planning.	
Pre-Conditions	The system has historical data on deployment reasons and external factors.	
Post-Conditions	The system identifies reasons for deployments and suggests next deployment areas.	
Main Success Scenario	User Action 1. The Fleet Manager requests an analysis of deployment reasons. 3. The Fleet Manager specifies areas of interest for analysis. 5. The Fleet Manager initiates the analysis process. 8. The Fleet Manager views the analysis results.	
	System Response 2. The system retrieves relevant historical data from the database. 4. The system filters the data based on specified areas. 6. The system analyzes the data to identify deployment reasons. 7. The system displays the reasons and suggests next deployment areas.	
Extensions	If no significant reasons are found, the system alerts the user with suggestions for further investigation.	

Table 7. Predict Stationing Days

UC-06	
Use Case Name	Predict Stationing Days
Scope	Predictive System for Commercial Shipping
Level	High-Level
Primary Actor	Fleet Manager
Stakeholder & Interest	Fleet Manager; interested in understanding how long vessels will be stationed for effective resource allocation.
Pre-Conditions	The system has historical data on vessel stationing durations.
Post-Conditions	The system provides predictions for future stationing durations based on historical data.
Main Success Scenario	User Action 1. The Fleet Manager requests predictions for stationing durations. 3. The Fleet Manager selects relevant factors (e.g., weather). 5. The Fleet Manager initiates the prediction process. 8. The Fleet Manager views the predicted durations.
	System Response 2. The system retrieves relevant historical data from the database. 4. The system analyzes the specified factors along with historical data. 6. The system analyzes the data to identify deployment reasons. 7. The system displays the predictions in a report format.
Extensions	If predictions indicate extended stationing durations, the system alerts the Fleet Manager with recommendations for resource allocation.

Table 8. Forecast Surge in Deployment

UC-07		
Use Case Name	Forecast Surge in Deployment	
Scope	Predictive System for Commercial Shipping	
Level	High-Level	
Primary Actor	Fleet Manager	
Stakeholder & Interest	Fleet Manager; interested in anticipating surges for optimal resource allocation.	
Pre-Conditions	The system has historical deployment data available for analysis.	
Post-Conditions	The system predicts potential surges in deployment based on the historical data.	
Main Success Scenario	User Action	System Response
	1. The Fleet Manager requests a surge prediction analysis.	2. The system retrieves historical deployment data from the database.
	3. The Fleet Manager specifies parameters for analysis (e.g., seasonality).	4. The system analyzes the data based on specified parameters.
	5. The Fleet Manager initiates the forecasting process.	6. The system predicts potential surges in deployments. 7. The system displays the forecasts in a detailed report format.
	8. The Fleet Manager views the surge predictions.	
Extensions	If surges are predicted, the system prompts the Fleet Manager with recommended actions for resource allocation.	

Table 9. Predict Transit Routes

UC-08		
Use Case Name	Predict Transit Routes	
Scope	Predictive System for Commercial Shipping	
Level	High-Level	
Primary Actor	Fleet Manager	
Stakeholder & Interest	Fleet Manager; interested in optimizing transit routes for efficiency.	
Pre-Conditions	The system has access to historical transit route data.	
Post-Conditions	The system generates predictions for probable transit routes based on historical movements.	
Main Success Scenario	User Action 1. The Fleet Manager requests transit route predictions.	System Response 2. The system retrieves relevant historical route data from the database.
	3. The Fleet Manager selects the parameters for analysis.	4. The system analyzes the specified parameters along with historical data.
	5. The Fleet Manager initiates the prediction process.	6. The system forecasts probable transit routes based on past movements.
		7. The system displays the predictions in a visual format (e.g., maps, graphs).
		8. The Fleet Manager views the predicted routes.
Extensions	If the predicted routes deviate significantly from previous patterns, the system alerts the Fleet Manager with suggested alternative routes.	

Table 10. Predict Duration of Deployment

UC-09		
Use Case Name	Predict Duration of Deployment	
Scope	Predictive System for Commercial Shipping	
Level	High-Level	
Primary Actor	Fleet Manager	
Stakeholder & Interest	Fleet Manager; interested in estimating deployment durations for planning purposes.	
Pre-Conditions	The system has historical data on deployment durations available.	
Post-Conditions	The system predicts expected durations of vessel deployments based on historical data.	
Main Success Scenario	User Action	System Response
	1. The Fleet Manager requests duration predictions for vessel deployments.	2. The system retrieves relevant historical deployment data from the database.
	3. The Fleet Manager specifies the parameters for the analysis (e.g., type of vessel).	4. The system analyzes the data based on the specified parameters.
	5. The Fleet Manager initiates the prediction process.	6. The system predicts expected durations of deployments based on historical trends.
		7. The system displays the predictions in a report format.
	8. The Fleet Manager views the predicted durations.	
Extensions	If the predicted duration exceeds certain thresholds, the system alerts the Fleet Manager with considerations for adjustments.	

Table 11. *Analyze Maintenance Data*

UC-10	
Use Case Name	Analyze Maintenance Data
Scope	Predictive System for Commercial Shipping
Level	High-Level
Primary Actor	Maintenance Team
Stakeholder & Interest	Maintenance Team; interested in understanding maintenance patterns to optimize scheduling.
Pre-Conditions	The system has historical maintenance data available for analysis.
Post-Conditions	The system identifies patterns and correlations in maintenance activities and conditions.
Main Success Scenario	User Action System Response
	1. The Maintenance Team requests an analysis of maintenance data. 2. The system retrieves relevant historical maintenance data from the database.
	3. The Maintenance Team specifies the criteria for analysis. 4. The system filters the data based on specified criteria.
	5. The Maintenance Team initiates the analysis process. 6. The system analyzes the data to identify maintenance patterns.
	7. The system displays patterns and correlations in a report format.
Extensions	8. The Maintenance Team views the analysis results.
	If no significant patterns are identified, the system alerts the user with recommendations for further data collection.

Table 12. Predict Future Maintenance

UC-11		
Use Case Name	Predict Future Maintenance	
Scope	Predictive System for Commercial Shipping	
Level	High-Level	
Primary Actor	Maintenance Team	
Stakeholder & Interest	Maintenance Team; interested in forecasting maintenance needs to reduce downtime.	
Pre-Conditions	The system has historical maintenance records and operational data.	
Post-Conditions	The system predicts future maintenance requirements for vessels based on historical data.	
Main Success Scenario	User Action	System Response
	1. The Maintenance Team requests maintenance predictions.	2. The system retrieves relevant historical maintenance data from the database.
	3. The Maintenance Team selects relevant vessels for analysis.	4. The system filters the data based on the selected vessels.
	5. The Maintenance Team initiates the prediction process.	6. The system predicts future maintenance needs based on the analysis.
		7. The system displays the predicted maintenance needs in a report format.
	8. The Maintenance Team views the predictions.	
Extensions	If the predictions indicate imminent maintenance needs, the system alerts the Maintenance Team with recommended actions.	

2.2 Functional Requirements

2.2.1 Module 1: Data Integration and Preprocessing Module:

- The system shall collect historical data from multiple sources, including AIS (Automatic Identification System) logs, radar data, and maintenance logs.
- The system shall be able to clean and preprocess the collected data to handle missing, incomplete, or erroneous data points.
- The system will integrate and unify the data into a single structured format for efficient analysis.

2.2.2 Module 2: Pattern Recognition and Correlation Analysis Module:

- The system shall identify patterns in deployment frequencies based on historical data, including peak deployment times and vessel type correlations.
- The system shall recognize correlations between operational statuses at various levels (e.g., unit, fleet) and external factors such as weather or sea conditions.
- The system shall visualize recognized patterns to assist decision-makers in understanding trends in fleet behavior.

2.2.3 Module 3: Predictive Analytics Module:

- The system shall develop predictive models to forecast future deployment patterns, including probable future surges in deployment frequency.
- The system will provide predictive insights on the number of assets that will be deployed based on vessel type, operational status, and historical usage trends.
- The system will forecast upcoming maintenance needs based on historical maintenance logs and external conditions such as weather.

2.2.4 Module 4: Anomaly Detection Module:

- It shall alert fleet managers in advance of upcoming required maintenance to prevent unexpected vessel downtime.
- The system shall identify and flag anomalies in vessel operation, maintenance schedules, or deployment patterns.
- The system shall notify fleet managers of any anomalies detected in vessel patterns, operational statuses, or maintenance schedules.

2.2.5 Module 5: User Interface and Visualization Module:

- The system shall provide a web-based interface for easy access to predictive analytics and historical trends.
- The system shall generate graphical reports (e.g., charts, graphs, heatmaps) that clearly depict predictive insights such as deployment trends, maintenance schedules, and operational risks.

- Users will be able to interact with the system through filters (e.g., by vessel type, time range) to customize their view of the data.

2.2.6 Module 6: System Integration and Testing Module:

- The system shall provide a testing module that allows for the validation of predictive models against historical datasets.
- The system will report on prediction accuracy, error rates, and thresholds to ensure the system's predictions meet practical operational needs.
- The system will provide detailed reports on system performance, accuracy, and areas of improvement.

2.3 Non-Functional Requirements

1. Scalability:

- Must handle large datasets and scale to accommodate multiple fleets data.

2. Performance:

- Near-time or In-time data processing with response times of less than 10 seconds.

3. Reliability and Availability:

- 99.9% system availability with fault tolerance and reliable backup mechanisms.

4. Usability:

- User-friendly interface requiring minimal training, with clear tutorials and documentation.

5. Accuracy:

- Predictive models should achieve an accuracy rate of 80% or higher.

6. Maintainability:

- Easy to maintain and update, with clear documentation and standard development practices.

7. Data Integrity:

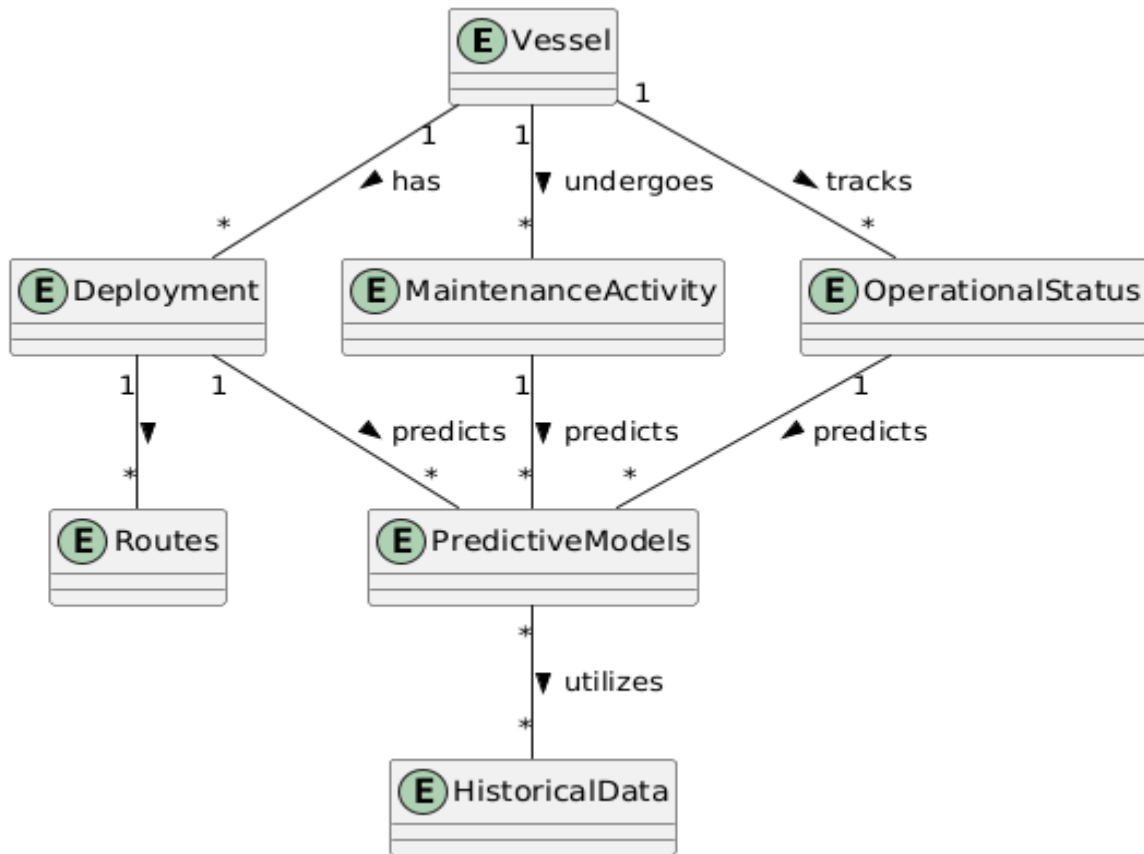
- Ensure accurate and consistent data processing without duplication or corruption.

8. Extensibility:

- Flexible architecture to allow future enhancements and integrations.

2.4 Domain Model

Figure 2. Domain Model

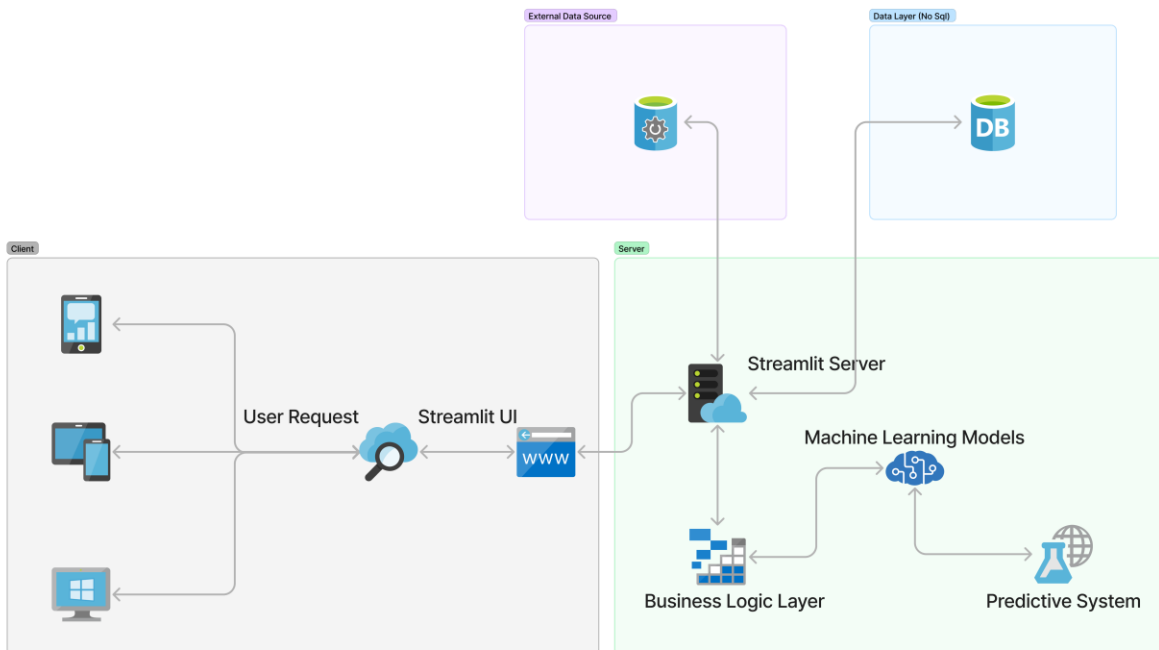


Chapter 3

3 System Overview

3.1 Architectural Design

Figure 3. *Architecture Diagram*



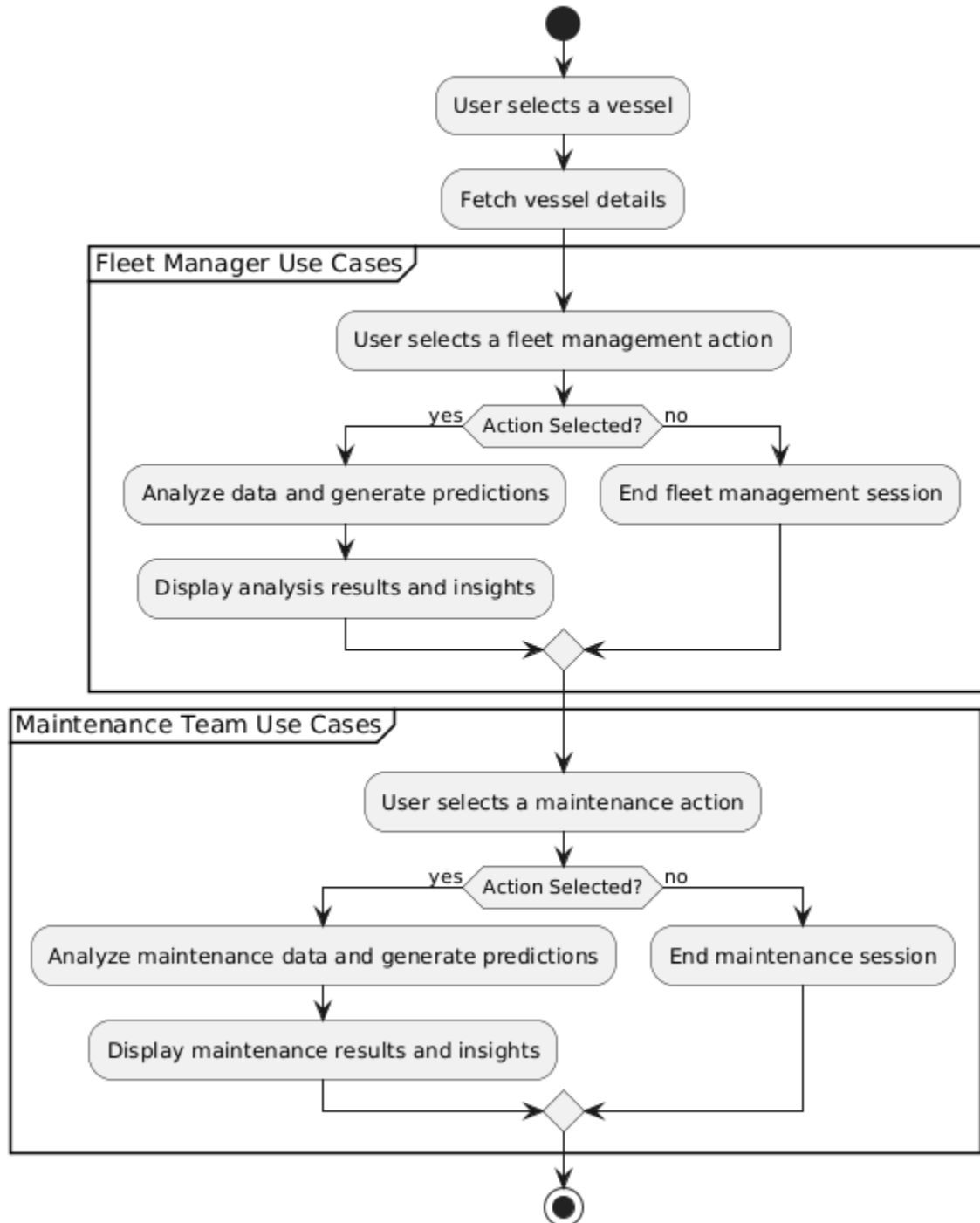
Interaction Flow:

1. **User interacts** with the web interface (client).
2. **Client sends request** to the Streamlit server (e.g., a user submits data or requests an analysis).
3. The **server processes** the request, retrieves or manipulates data, possibly makes predictions using models, and sends results back.
4. **Server responds** with processed data or visualizations, and the client updates the interface accordingly.

3.2 Design Models

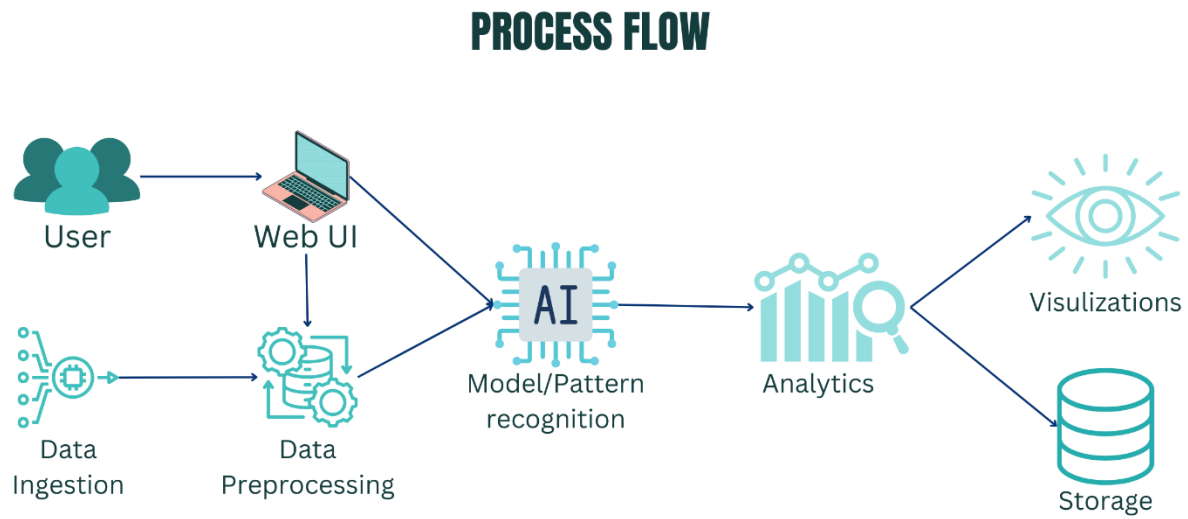
3.2.1 Activity Diagram

Figure 4. Activity Diagram



3.2.2 Process Flow

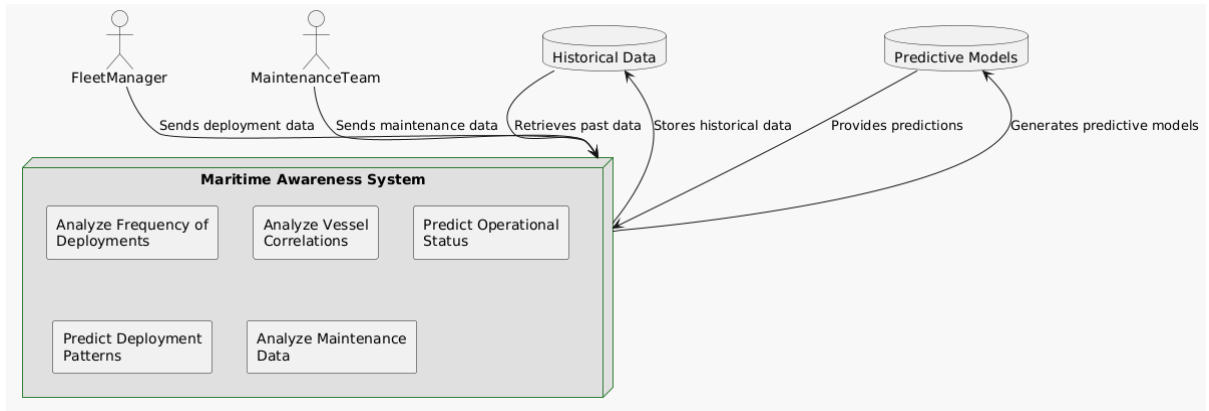
Figure 5. Process Flow



3.2.3 Data Flow Diagram

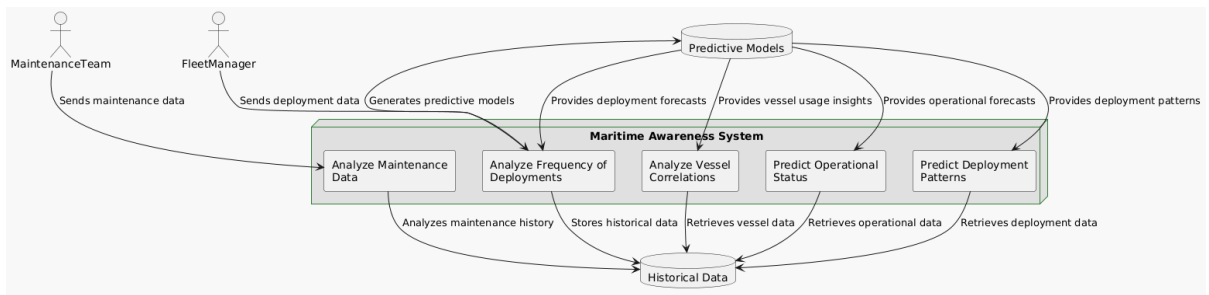
DFD Level 0:

Figure 6. DFD Level 0



DFD Level 1:

Figure 7. DFD Level 1



3.2.4 System Sequence Diagrams

Figure 8. *Analyze Frequency of Deployments*

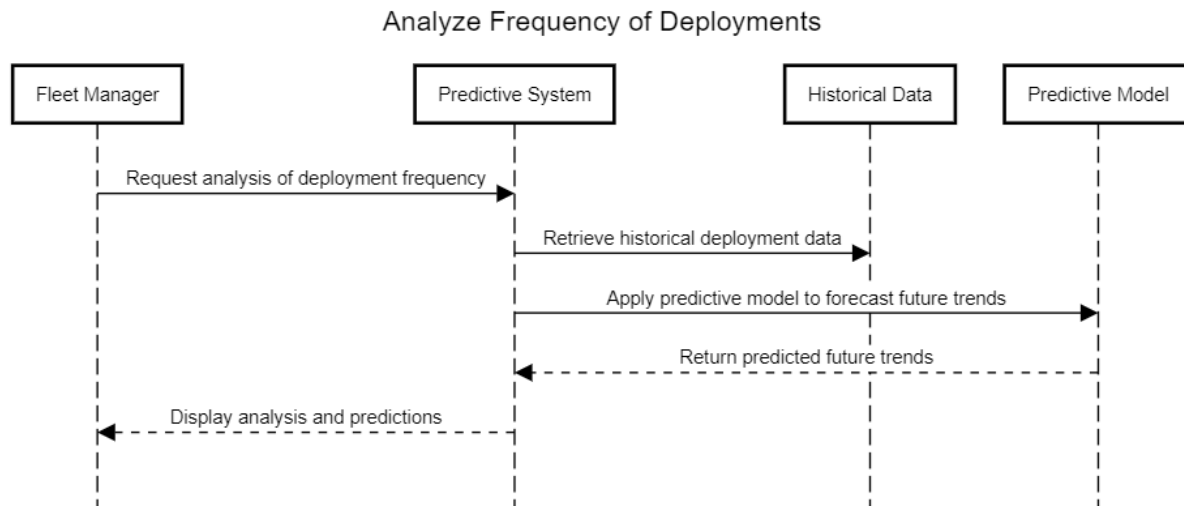


Figure 9. *Analyze Reasons for Deployment*

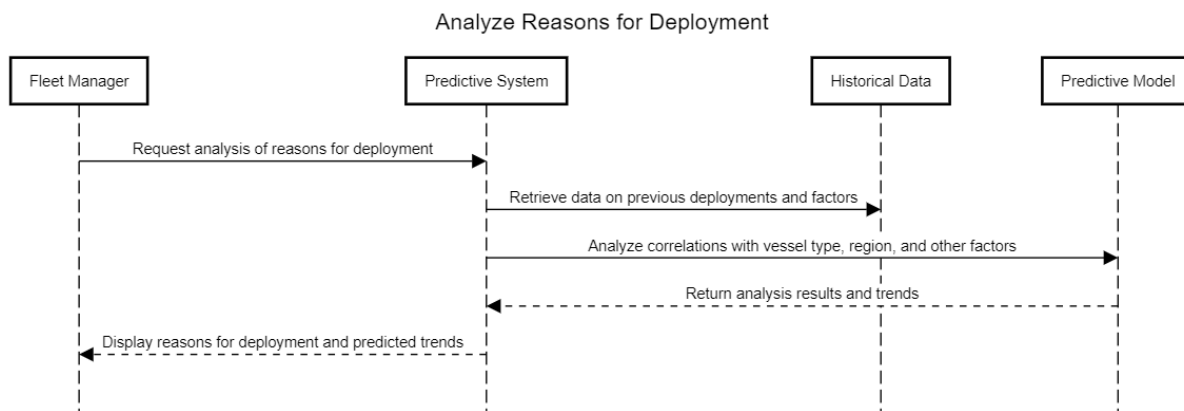


Figure 10. *Analyze Vessel Correlations*

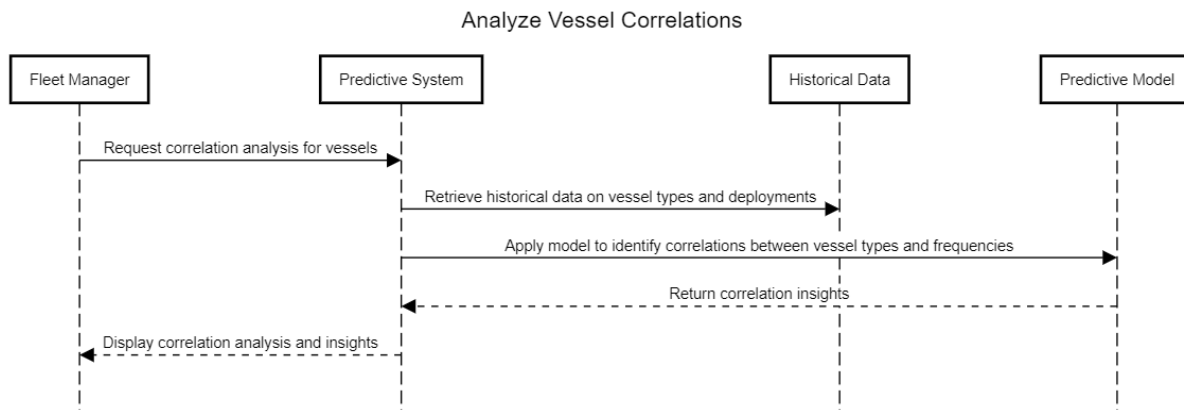


Figure 11. Forecast Surge in Deployment

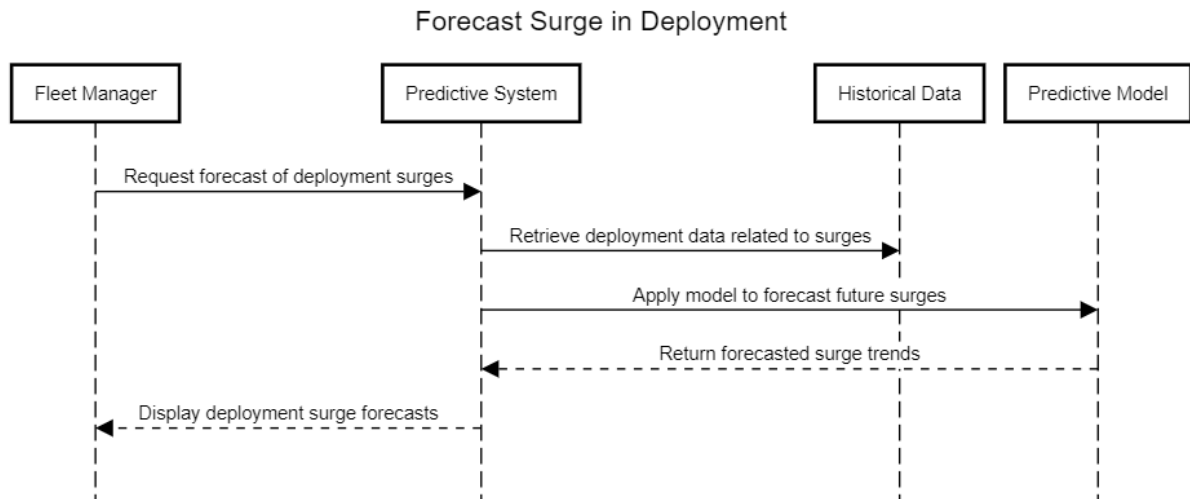


Figure 12. Predict Deployment Patterns

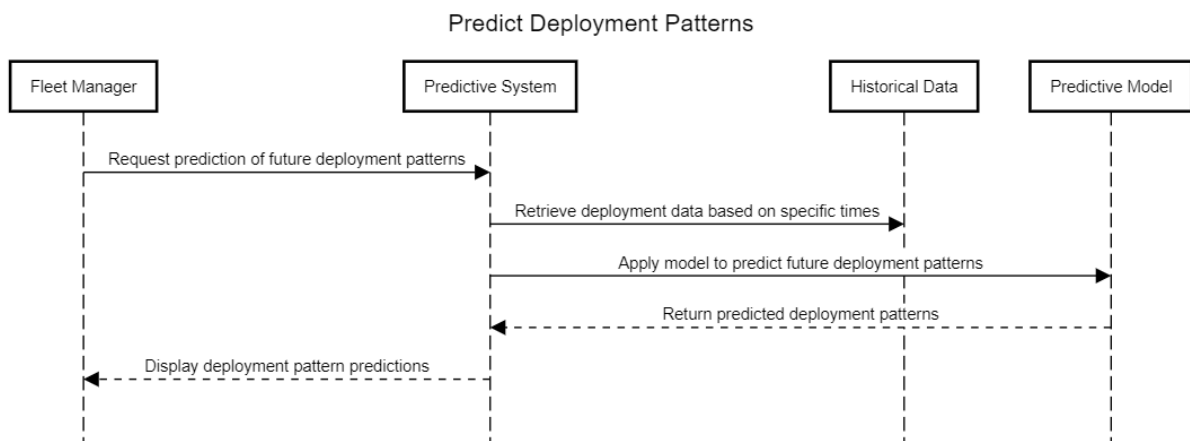


Figure 13. Predict Duration of Deployment

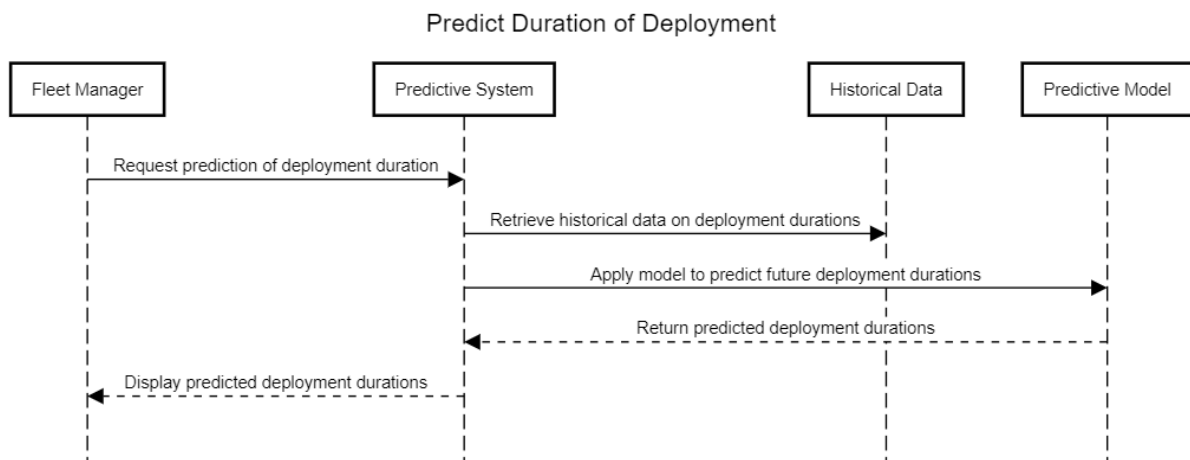


Figure 14. Predict Future Maintenance

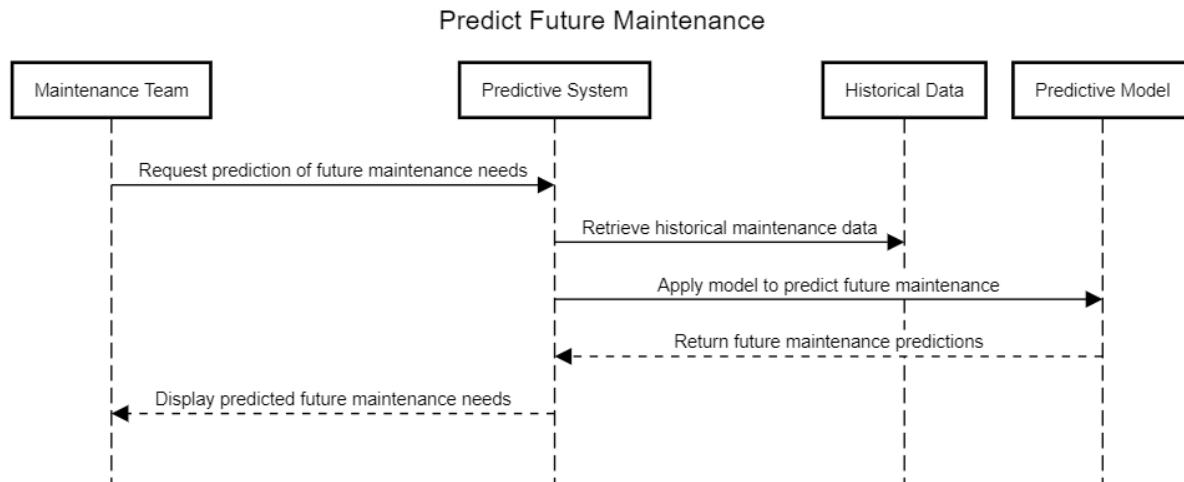


Figure 15. Predict Operational Status

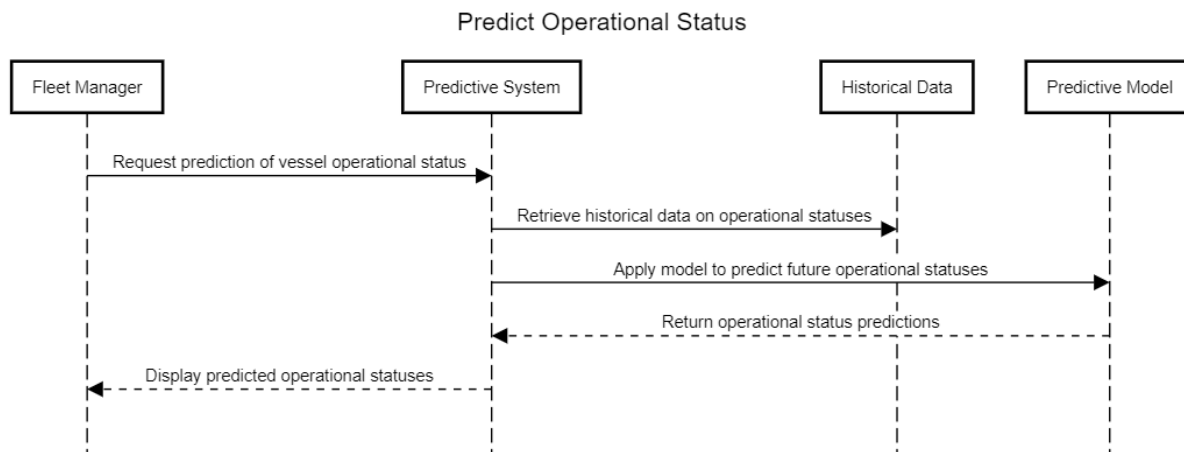


Figure 16. Predict Stationing Days

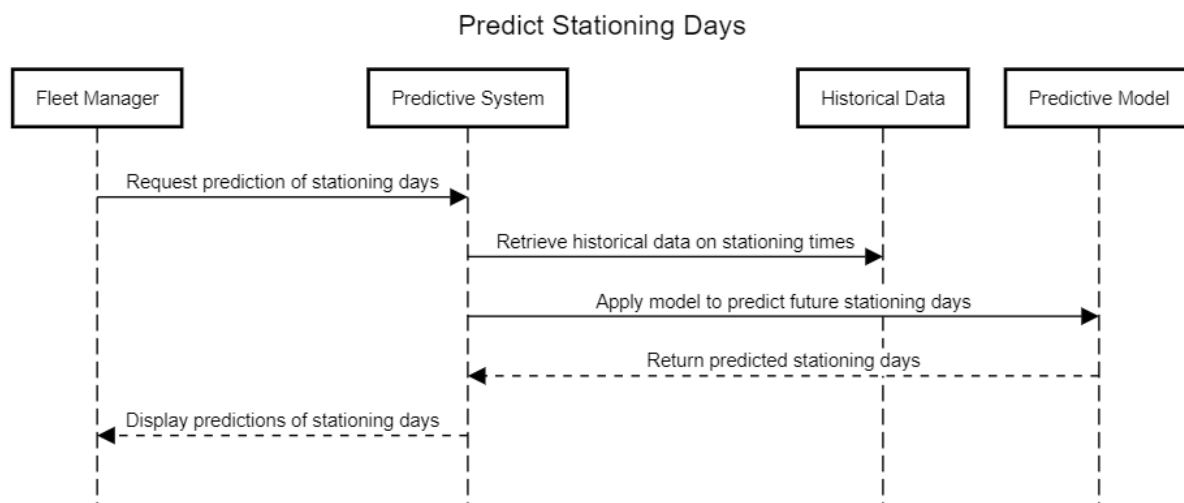


Figure 17. Predict Transit Routes

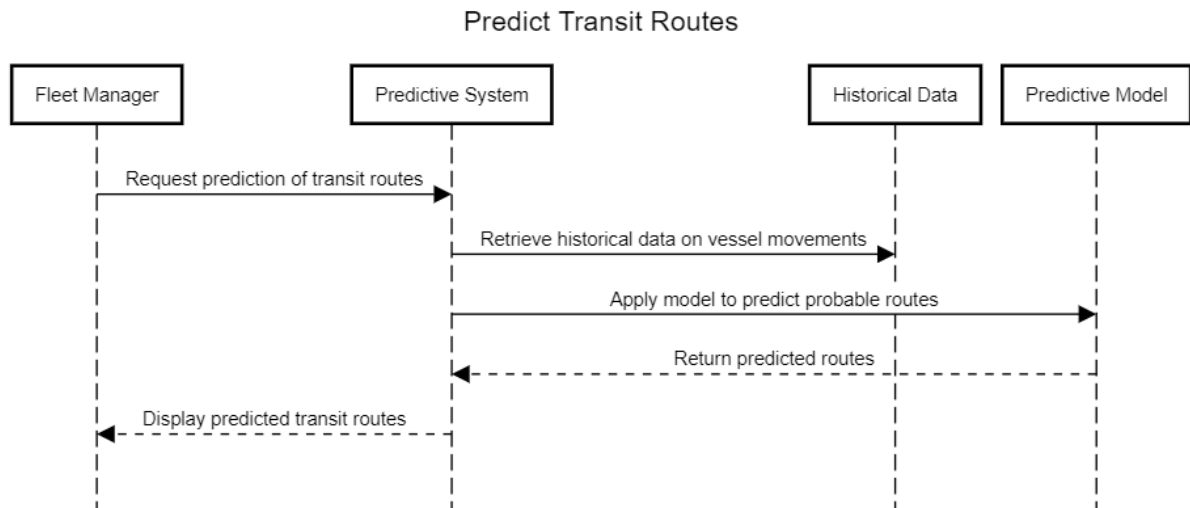
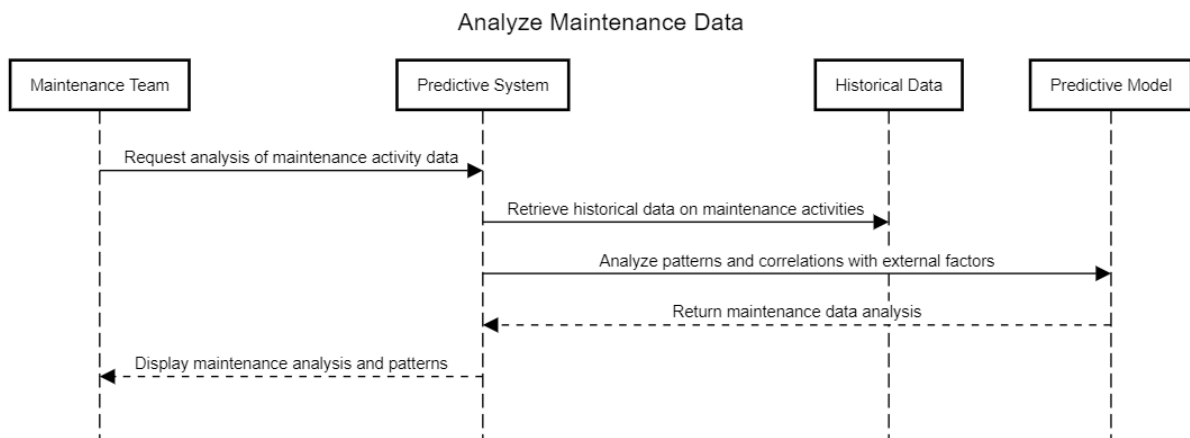
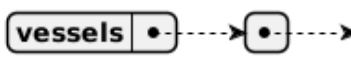


Figure 18. Analyze Maintenance Data



3.3 Data Design

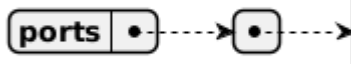
Figure 19. Vessel AIS Data



The diagram shows a rounded rectangle labeled 'vessels' with a small circle inside. A dashed arrow points from this circle to another small circle, which then points to the first column of the table below.

timestamp	02/04/2024 00:00:00
type_of_mobile	Class A
MMSI	219543000
latitude	91.000000
longitude	0.000000
navigational_status	Unknown value
ROT	□
SOG	□
COG	□
heading	□
IMO	9944546
callsign	OZGV2
name	LAURA MAERSK
ship_type	Cargo
cargo_type	No additional information
width	32
length	172
type_of_position_fixing_device	Combined GPS/GLONASS
draught	7.6
destination	SEHAD
ETA	03/04/2024 04:00:00
data_source_type	AIS
size_A	154
size_B	18
size_C	1
size_D	31

Figure 20. Worlds Port Data



The diagram shows a rounded rectangle labeled 'ports' with a small circle inside. A dashed arrow points from this circle to another small circle, which then points to the first column of the table below.

INDEX_NO	8895.0
REGION_NO	8650.0
PORT_NAME	CONVENT
COUNTRY	US
LATITUDE	30.016667
LONGITUDE	-90.833333

Table 13. Vessel AIS Data

Attribute	Data Type	Description
timestamp	String	Timestamp from the AIS basestation
type_of_mobile	String	Type of target this message is received from
MMSI	Integer	MMSI number of vessel
latitude	Float	Latitude of message report
longitude	Float	Longitude of message report
navigational_status	String	Navigational status from AIS message
ROT	Float	Rate of turn from AIS message
SOG	Float	Speed over ground from AIS message
COG	Float	Course over ground from AIS message
heading	Integer	Heading from AIS message
IMO	Integer/String	IMO number of the vessel
callsign	String	Callsign of the vessel
name	String	Name of the vessel
ship_type	String	AIS ship type of this vessel
cargo_type	String	Type of cargo from the AIS message
width	Integer	Width of the vessel
length	Integer	Length of the vessel
type_of_position_fixing_device	String	Type of positional fixing device
draught	Float	Draught field from AIS message
destination	String	Destination from AIS message
ETA	String	Estimated Time of Arrival
data_source_type	String	Data source type, e.g. AIS
size_A	Integer	Length from GPS to the bow
size_B	Integer	Length from GPS to the stern
size_C	Integer	Length from GPS to starboard side
size_D	Integer	Length from GPS to port side

Table 14. *Worlds Port Data*

Attribute	Data Type	Description
INDEX_NO	Float	Unique identifier for the port
REGION_NO	Float	Region number the port belongs to
PORT_NAME	String	Name of the port
COUNTRY	String	Country code where the port is located
LATITUDE	Float	Latitude coordinate of the port
LONGITUDE	Float	Longitude coordinate of the port

References

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